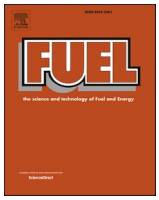
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Full Length Article



Experimental investigation of the impact of CeO_2 nanoparticles in Jet-A and Jatropha-SPK blended fuel in an aircraft can-combustor at flight conditions

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ABSTRACT

International consensus aimed at mitigating climate change, global perspective shift towards clean energy, uncertainties in conventional fuel supply, along with increasingly stringent emission norms for the aviation sector has prompted the ongoing research on low emissions aviation fuel technology. Consequently, aviation fuels blended with synthetic paraffinic kerosenes (SPKs) derived from Jatropha biomass have been tested extensively. The effect of nanoparticles (NPs) in the fuel blends towards emission reduction is not studied for aero-engines. This study aims to investigate the emissions impact of Jatropha SPK blended Jet-A aviation fuel along with CeO_2 (Ceria) NPs, in an aircraft engine can-combustor operated at representative flight conditions. The base fuel blends used were Jet-A (ATF), 80% Jet-A with 20% Jatropha-SPK (A80J20), and 30% Jet-A with 70% Jatropha-SPK (A30J70). NPs at 0.5% and 1% weight concentration were added to the base fuel blends, which resulted in nine different fuel types. NOx reduction is observed with increasing CeO_2 NP concentration. CO, UHC and soot reduction occurs with increasing Jatropha and CeO_2 content. CeO_2 NPs reduce all the emissions of concern, and could be an important additive for aviation fuel.

1. Introduction

Forecasts for the air transport sector predict air traffic increase by a factor of two in the next twenty years, along with a steady increase in oil demand at the rate of million barrels per day in the coming years [1,2]. The demand for aviation fuel is expected to outpace the supply. Conventional fossil fuels use is considered a major challenge for the 'Net Zero by 2050' initiative, which is a roadmap for change in global energy consumption pattern towards net zero emissions, put forth by the International Energy Agency (IEA). An important step towards fulfilling the roadmap is reducing dependence on fossil fuels. Biofuels are important for their potential to lower green-house gas emissions and can be a good substitute for fossil fuels [3,4]. In the aviation sector, emission standards have become increasingly stringent across the globe [5]. This is partly due to the legally binding framework for limiting global average temperature increase to within 1.5–2 °C above pre-industrial levels, as agreed upon in the Paris Agreement [6]. Due to the global effort for mitigating climate change by moving towards a zero emissions and sustainable society, there is considerable amount of contemporary research in aviation biofuels [7,8].

Nanoparticles (NPs) have size ranging from 1 to 100 nm, and they are manufactured from metals, metal oxides, or other inorganic substances. NPs are manufactured from pure metals (or base metals) or their compounds (usually oxides), by the process of modifying the larger macroscopic structure of the base metal or its compound to a structure consisting of nano-scale particles or nanoparticles. "Nanoparticles assist in the atomization, evaporation and combustion process in several ways. NP addition results in lower blend viscosity, higher cetane number, therefore reduced ignition delay, and better heat transfer to fuel droplets. Thus, nanoparticles favor fuel atomization, penetration and thermal energy transport. These factors help to reduce the ignition delay period and hence enhanced combustion characteristics are observed in the presence of nanoparticles in the fuel" [22].

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Nomenclature

t	Time, s
u_i	Velocity component, m/s
S	Source term
M	Molecular weight
c_p	Constant-pressure Specific heat, J/kg-K
Y	Species mass fraction
Subscripts	
i	x, y, or z direction
nf	Nanofuel (Fuel + NP)
NP	Nanoparticle
Greek	
ρ	Density
Φ	Diffusion parameter
Γ	Effective diffusion coefficient
ϕ	Nanoparticles volume fraction

For lower than 5–10 nm particles, the ratio of NP surface area compared to the base metal/compound surface area can increase by up to 90% with very different properties compared to the base metal/compound. Due to the large surface area provided by NPs for chemical reactions, their addition to liquid fuels increases the rate of reaction during combustion.

The high energy density of NPs makes them particularly suitable for aerospace and propulsion applications [9]. Besides, NP additives reduce the ignition delay [10], assist in quicker and more complete combustion [11], reduce emissions and brake specific fuel consumption (BSFC) in internal combustion engines powered by diesel and/or biodiesel [12,13]. In addition, they are beneficial to thermal efficiency, and have enhanced catalytic activity [14]. There is considerable amount of literature relating to NP additives in biofuels and conventional fuels (diesel, gasoline, aviation fuel). Some advantages and disadvantages of NPs addition to fuels is shown in Table 1.

Some of the representative works relating to NP additives in diesel/biodiesel are mentioned. Soudagar et al. [12] reviewed nano-additives in diesel–biodiesel blends. Reduction of hydrocarbons (HC), carbon monoxide (CO) and particulate emissions with NP additives were reported and some of the literature is discussed in the following lines. Furthermore, biodiesels in general have higher BSFC and lower BTE, which can be reversed by metallic NP additives on account of higher calorific value, catalytic activity and better combustion. Gad et al. [15] experimented with diesel engine performance and emissions using 80% diesel-20% biodiesel blend, with carbon nano-tube (CNT) and carbon nano-sheet (CNS) additives ranging from 25 to 100 ppm. Improvement in BTE, THC, nitric oxides (NOx) and reduced ignition delay was observed. Additionally, improvements with 100 ppm CNS were almost twice that with 100 ppm CNT for the mentioned parameters. Waheed [16] performed a numerical study using Al_2O_3 nanoparticles in generic aviation turbine fuel (ATF) for diffusion combustion. Thermal conductivity, heat capacity and viscosity of nanofluid fuel were higher whereas activation energy was lower. Nanofluid fuel was consumed more quickly compared to base fuel and also generated higher temperatures. Khond et al. [17] reviewed the performance of CI engines with NP additives to biodiesel/diesel along with the effect on fuel properties. NPs facilitate catalytic activity due to large surface area to volume ratio and better combustion due to micro-explosion phenomenon. Mehta et al. [18] experimented with sonicated aluminium (Al), iron (Fe) and boron (B) nanoparticles combined with diesel, in a CI engine. Higher in-cylinder temperatures were noted for NPs (Al, Fe and B, in that order) in the fuel which aided in better BTE. Fall in CO and THC, with Al and Fe nanofuel, and slight rise in NOx was detected. Fuels with NP were reported to have better combustion characteristics due to higher energy density, catalytic activity, and burning and heat release rates, combined with lower ignition delay. Ooi et al. [19] assessed diesel engine performance and emission characteristics with graphite oxide (GO), single walled carbon nanotubes (SWCNT), and CeO_2 (Ceria) additives to diesel fuel. SWCNT at 25 ppm was most effective in reducing ignition delay, BSFC, CO and THC, after GO; cerium additive based fuel was less effective and economical. However, NOx was found to increase with SWCNT (15%) and GO (9%) due to higher combustion temperatures.

Khan et al. [20] reviewed nanoparticles produced from metals and metal oxides and the combustion characteristics of diesels with addition of nanoparticles. They reviewed literature and found that nanoparticles owing to the high surface area provide better combustion characteristics, high reactivity, shortened ignition delay and fast energy release. They also showed that Nanoparticles of CeO_2 NP are more effective in NO reduction due to their unique catalytic properties.

Tomar et al. [21] reviewed the influence of nanoadditives on emission characteristics of Compression Ignition engines fuelled with diesel, biodiesel, and blends. They summarized that NP blended to base fuel increases surface to volume ratio, and heightens the heat and mass transfer. They also improve the thermal efficiency and specific fuel consumption of the diesel engine. They further highlight the improvement in injection timing, and enhancement in properties such as viscosity, calorific value, flash point, and fire point with NP concentration.

Kumar et al. [22] demonstrated that 80 ppm CeO_2 NP with up to 40% oxygenated biodiesels form waste cooking oil blended with neat diesel showed emissions reduction of up to 16%, 6%, and 12% respectively for UHC, soot, and NOx. The maximum reductions were obtained for the blends with a maximum percentage of biodiesel along with CeO_2 NPs.

Literature concerning aviation fuels blended with biofuels and/or nanoparticles is now mentioned. Booma devi et al. [23] reported up to 94% reduction in NOx with 3% TiO_2 NPs in aviation fuel blended with butanol, jatropha methyl ester, soya methyl ester and rapeseed methyl ester. The results were applicable for a micro gas turbine and could be useful for aviation gas turbines. Kannaiyan et al. [24] investigated the influence of nanoparticles' dispersion on the physical properties and spray performance of aviation fuel (Jet-A1) with alumina nanoparticles. Higher nanoparticle concentration of 4% by weight correlated with higher fuel viscosity (up to 2%) and density, while surface tension was found to decrease (by up to 3%), compared to base fuel. The liquid sheet breakup length decreased with increasing nanoparticle concentrations. Also, the mean droplet diameters of NP blended fuel were lower than the base fuel. Ferrao et al. [25] studied the combustion characteristics of a hydroprocessed vegetable oil (HVO) drop blended with Aluminum (Al) NPs of 40 nm and 70 nm at weight concentrations of 0.5 and 1%. They suggest that combustion characteristics of pure HVO could be enhanced with the addition of Al NPs. Lower NP size resulted in a slight increase in the burning rate of nanofuels, whereas increase in the NP concentration lead to significantly higher burning rate, lower ignition delay and also heightened intensity and lower delay of micro-explosions, a "disruptive burning phenomena" that occurs at the end of a droplet lifetime.

Aviation fuels differ in properties from diesel/gasoline in terms of properties and have stringent specifications [26,27,28]. ATF is a generic

Table 1
Advantages and Issues with NPs added to base fuels. [12,17,20,21,24,25].

Advantages	Issues
Reduce Ignition Delay, High Surface to volume ratio, Catalytic activity, Increase cetane number, Assist in droplet break up by mocro-explosion, Disruptive burning, Enhance combustion efficiency, Reduce CO, UHC Soot compared to base fuel. High Energy Density, Desirable for propulsion applications.	Miscibility/Solubility in liquid fuels, Heterogeneous combustion kinetics, Formation of unoxidized and unstable nanoparticles fuel suspension, Native oxide passivation, Reactivity for accelerating ignition, Cost and scalability.

aviation fuel, and it may refer more specifically to Jet-A, Jet-A1, JP-5, JP-8 etc. Any further reference to ATF in the following text would mean Jet-A.

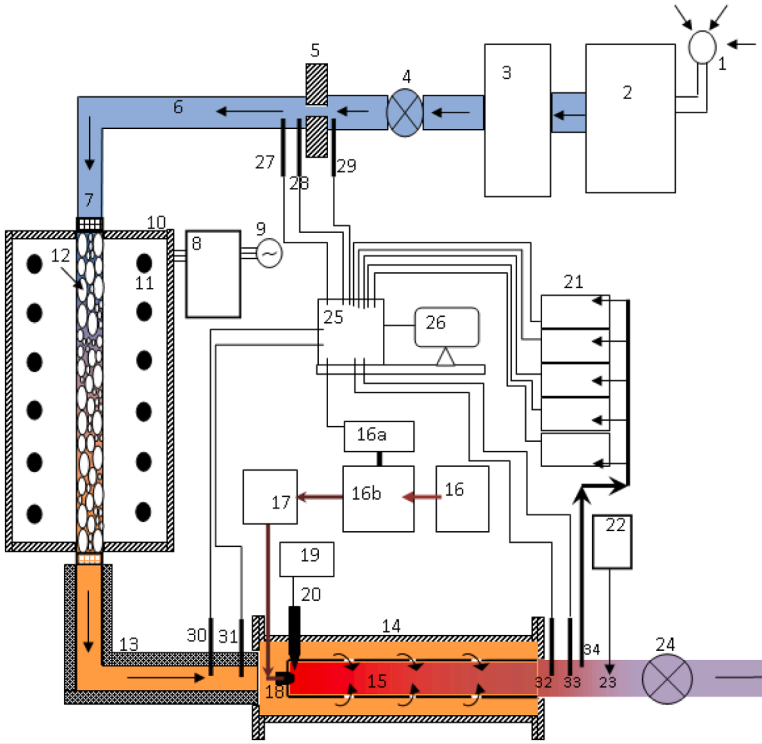
Jatropha Oil to jet (OTJ) SPKs from conventional hydro-processing route can provide 75% GHG reduction compared to Jet-A at the time of fuel production. Jatropha biojet made from hydroprocessing has very little amount of aromatics [37], whereas the Jatropha produced from catalytic hydrothermolysis has upto 10.8% by weight aromatics content [38]. However, this does not limit the use of Jatropha since the minimum aromatic quantity required in aviation fuels is 8% as per ASTM D7566 [38], and the aromatic content is sufficiently available in Jet-A. Also, compared to Camelina, Jatropha yields more oil per unit cultivated area and is about four times cheaper than Camelina oils. It is therefore the most widely used biomass source for SPKs [37].

NPs are generally of interest in aviation sector and aircraft propulsion due to their high energy density [29]. The literature concerning aviation fuel blended with NPs is generally related to laboratory-specific applications/devices or fundamental studies relating to fuel characteristics [30,31,32,33]. Therefore, there appears a gap in the literature relating to emissions from aircraft gas turbine combustors, ***using*** NP blended aviation fuel, aviation biofuel or biojet, and representing actual flight conditions.

Various nanoparticle types such as CeO_2, Al_2O_3, etc. have been used for engine emissions investigations. Most of the researchers have found a decrease in CO, UHC and soot emissions with the use of Al_2O_3 and CeO_2 nanoparticle in fuel; however, only CeO_2 has been known to reduce NOx emissions. "Al_2O_3 NPs enhance the combustions by atomization, dispersion and effective heat transfer between the fuel droplets. But there is no catalytic activity to reduce emissions. Though the Al_2O_3 NPs are cheaper, they produce more NOx compared to neat fuel combustion without NPs. This limits the use of Al_2O_3 NPs in engine applications." [22]. Since the catalytic activity that reduces NOx emissions is seen only by the action of CeO_2 NPs in the literature, we have chosen to use these in the study. An aircraft engine can-combustor supplied by Pratt and Whitney (PWC) is tested for emission characteristics from the combustor by using CeO_2 NP blended ATF-Jatropha-Biojet (also supplied by PWC). The tests were carried out for representative flight conditions.

2. Methodology

This section is divided into three parts, namely, the Experimental Details and Procedure in Section 2.1, Fuel Compositions description in Section 2.2, and Nanoparticle characterization in Section 2.3. The experimental schematic, procedure and operating condition details are described in the first of these sections. In the succeeding section, there is a description of the biofuel blends used and their characteristics. The last section (Section 2.3) deals with Cerium oxide (ceria or CeO_2) nanoparticle characterization. All the fuels used along with NP additives and also the can-combustor were supplied by PWC. The experimental setup was built at Combustion Laboratory in KSU. Emission analyzer, Coriolis flow meter, Thermocouples, Pressure Transducers etc. were purchased from vendors. The 200 kW heater was designed by our team and manufactured by a private contractor.



1. Air intake, 2. Air compressor powered by diesel generator, 3. Air storage tank, 4. Inlet control valve, 5. Orifice plate, 6. Pipe entering into furnace, 7. Wire mesh, 8. Control panel for furnace, 9. Three phase electrical supply, 10. Furnace chamber, 11. Cylindrical rod elements for heating, 12. Ceramic balls, 13. Furnace outlet pipe insulation, 14. Can combustor, 15. Flame tube of combustor, 16. Fuel tank, 16a. Motor, 16b. Fuel pump, 17. Coriolis flow meter, 18. Atomizer, 19. Battery, 20. Ignitor, 21. Emission analyser, 22. High pressure water pump, 23. High pressure water spray for cooling, 24. Butterfly valve, 25. Data acquisition system, 26. Personal computer, 27. Temperature sensor, 28. Pressure sensor after orifice plate, 29. Pressure sensor before orifice plate, 30, 31. Temperature, Pressure sensors prior to combustor, 32. Five probe temperature sensor at combustor exit, 33. Pressure Sensor at combustor exit, 34. Sampling point for emission analysis, 35. Exhaust gas exit

Fig. 1. Schematic of experimental test rig.

2.1. Experimental details and procedure

The schematic of the apparatus used for assessing the emission characteristics due to the combustion of JetA-Biofuel-NP fuel blends within an aircraft engine can-combustor is shown in Fig. 1. A 20 bar capacity screw compressor supplies compresses air to a storage tank, which then supplies air to the system through a control valve. An Orifice flow meter is used to measure the air flow rate. A wire mesh is fixed at the mouth of a 200 kW electric furnace with 24 vertical heating elements of sintered carbide. The pipe section within the furnace is filled with ceramic balls in order to increase the heat capacity and act as a thermal reserve to adjust quickly to the fluctuations in operating conditions. The pipe is insulated in order to avoid heat losses. The high pressure, heated air, enters the can-combustor, where the fuel supply coming through a coriolis flow meter is atomized and the air–fuel mixture is ignited. For a gas turbine combustor, temperature and pressure at inlet and exit of the combustor are denoted as T3, P3 and T4, P4, respectively. Five probe thermocouples are used to acquire temperature signals T3 and T4. Piezoelectric pressure transducers are used to measure the pressure at relevant points, including P3 and P4. A high pressure water pump is used to supply water to cool the exhaust prior to an electronically controlled butterfly valve that is used to maintain pressure within the system. The exhaust is sample and fed to an emission analyzer. The entire setup is instrumented by National Instruments (NI) LABVIEW® Data Acquisition (DAQ) device. Fig. 2 (Supplementary Section) shows an infrared image of the fired-up can-combustor. The experimental test conditions are detailed in Table 2. The exhaust gases are analyzed using a multi-modular system from SIGNAL® instruments. The specifications of emission analyzer modules are given in Table 3 (Supplementary Section).

2.2. Fuel compositions

The base fuel used in the study is Jet-A, which is specific type of aviation turbine fuel (ATF). It is a crude oil distillate in the temperature range 478–533 Kelvin, and composed of saturated aliphatics such n-alkanes and *iso*-alkanes, naphthalenes and aromatics [34].

The fuel carbon number range for generic ATF is 8 to 16. Alkanes in the fuel provide high H/C ratio, which increases fuel energy density, naphthalenes reduce freeze point and aromatics aids in prevention of leakage at seals, but are also responsible for higher soot formation and smoke. However, recent research suggests that naphthalenes are more effective soot precursors compared to aromatics [35]. Biojets are bio-derived jet fuels or aviation fuels. Their chemical composition is similar to kerosene or diesel fuel, and 90% of the carbon species have carbon number ranging from 9 to 14. The majority of biojets are synthetic paraffinic kerosenes (SPKs) derived from Jatropha and Camelina, among other sources. Their chemical composition is closer to kerosene, therefore the term kerosene in SPK.

Henceforth, we may use the terms SPK or biojet interchangeably. SPKs also contribute towards lower emissions, have good thermal stability, good cold flow characteristics (e.g, low cloud point) and higher cetane number (therefore, lower ignition delay) [36].

Selected thermo-physical of Jet-A, Jatropha-SPK and Camelina-SPK are shown in Table 4. Composition and properties of the fuel blends (without NP) are shown in Table 5. Fuel blend compositions with CeO_2 NPs are shown in Table 6.Table 7

"Apart from pure ATF (Jet-A), the blend percentages provided by Pratt and Whitney involves two blends, one with higher (70%) Jatropha concentration (A30J70), and the other with lower (20%) Jatropha concentration (A80J20).

Lower Jatropha concentrations (upto 50%) have used in number of flight tests [38], and also approved by ASTM D7566 specifications. The reason for using higher Jatropha blend is that the aircraft engine manufacturer and our funding partner (PWC) wants to test higher Jatropha blends in their engine.

Table 2
Test condition details.

Condition	C1	C2	C3	C4	C5
Description	**Minimum Idle**	**Flight Idle**	**Cruise**	**Descent/ Climb**	**Land/ Take-off**
Percent Power	7–10	30–35	60–65	80–85	90–100
Overall Equivalence Ratio	0.040	0.121	0.227	0.273	0.318
Primary Zone Equivalence Ratio	0.12	0.605	1.135	1.365	1.59
P3 (gauge) , MPa	0.2466	0.346	0.588	0.852	0.931
T3 , Kelvin	390	510	580	625	650

Table 3
Emission analyzer modules details.

Module/Make	Range and Accuracy	Operating Principle
Total hydro carbon analyzer; 3000 HM	0–10000 ppm/ ±1% range or±0.2 ppm whichever is greater	Flame Ionisation Detector
NOx Analyser; 4000VM	0–10000 ppm (NO, NO_2 andN_2O)/ ±1% range or ± 0.2 ppmwhichever is greater	Chemiluminescence
Multigas Analyser; 9000 MGA	0–5% vol (CO); 0–25% vol (CO_2); 0–25% vol (O_2)	Paramagnetic (for O_2) Infrared (for CO and CO_2)

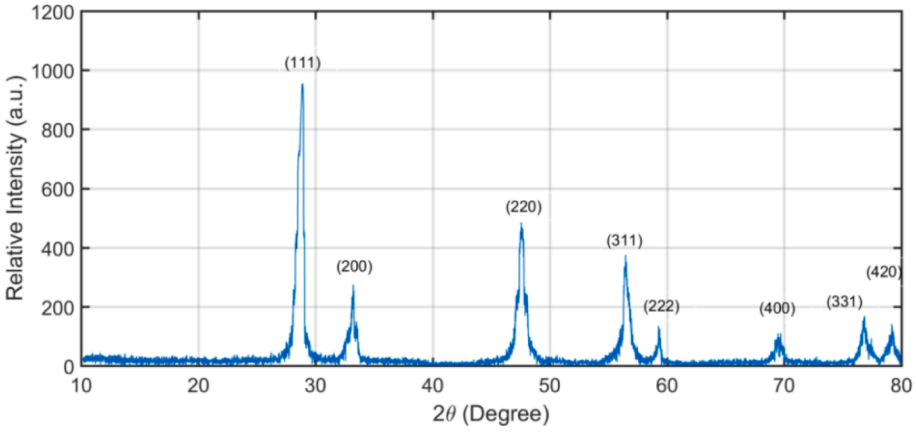


Fig. 2. Infrared image of combustor during operation.

Table 4
Standard Specifications of Jet-A, Jatropha SPK and Camelina SPK; adapted [36–39].

Property	ASTM Method	Jet A-Standard ASTM D1655 specifications	Jatropha SPK	Camelina SPK
Aromatics (%)	D1319	25 (max)	0.0	0.3
Naphthalenes (%)	D1840	3 (max)	0.12	<0.01
Smoke Point (mm)	D1322	25 (min)	53.0	50.0
Flash Point (°C)	D3828	38 (min)	46.5	42
Lower Heating Value (MJ/kg)	D4529	43.15	44.3	44.1
Viscosity@-20 °C (mm^2/s)	D445	8.0 (max)	3.663	3.336
Lubricity (mm)	D5001	0.85	0.76	0.76
Hydrogen (% mass)	D3345	13.96	15.5	15.1
Density@15 °C (kg/m^3)	D4052	775–840	749	753
Initial Boiling Point (°C)	D86/ D2887	165	164.5	162
Final Boiling Point (°C)	D86/ D2887	265	254.9	251.2
Freeze Point (°C)	D2386	−40 (max)	<-54.5	<-63.5
Sulphur (% mass)	D1266/ D5453	0.03	0.0001	<0.0001
H/C ratio	–	1.952	2.1875	2.1785
Empirical Formula	–	$C_{12.5}H_{24.4}$	$C_{11.2}H_{24.5}$	$C_{11.2}H_{24.4}$

Table 5
Properties of fuel blends used in the study.

BLEND NAME	ATF	A80J20	A30J70
Jet-A (vol%)	100	80	30
Jatropha (vol%)	0	20	70
COMPOSITION			
Aromatics	19.7	18.3	14.1
Density@15 °C (kg/m^3)	812.7	804.8	785.3
Viscosity@-20 °C (mm^2/s)	4.55	4.15	3.18
Freezing Point (°C)	−53.4	−55.9	−60.4
Net Heat of Combustion (MJ/kg)	42.91	42.99	43.22
Flash Point (°C)	51	40	39.5
Hydrogen (mass%)	13.67	13.83	14.24
Smoke Point (mm)	20.5	22	23
Naphthalenes (vol%)	1.62	1.32	0.61

Table 6
Fuel Blends with CeO_2 NPs.

ATF	ATF + CeO_2 (0.5% by wt.)	ATF + CeO_2 (1.0% by wt.)
A80J20	A80J20 + CeO_2 (0.5% by wt.)	A80J20 + CeO_2 (1.0% by wt.)
A30J70	A30J70 + CeO_2 (0.5% by wt.)	A30J70 + CeO_2 (1.0% by wt.)

Table 7
Percent NOx change with respect to ATF at maximum power.

	ATF	A80J20	A30J70
No Ceria	0	9.56	13.04
0.5% Ceria	–23.48	−16.08	−13.52
1.0% Ceria	−38.57	−30.43	−28.26

There have been very few airlines that have done testing with more than 50% biofuel blends [38], the effect of higher biofuel blend percent needs to be tested and this has been done with the PWC supplied fuel blends and can-combustor."

2.3. Ceria characterization

CeO_2 (Ceria or cerium oxide) NPs are added to Jet-A-Biofuel blends, and the NP type, concentration and fuel blends were supplied by Pratt and Whitney (PWC), Canada.

"The use of CeO_2 nanoparticles is preferred assist in the reduction of NOx as well as other emissions of concern; this is not found for most other, where the reduction of NOx is not observed. This is attributed to the characteristic redox reaction due to CeO2 that assists in the reduction of NOx (and also CO, UHC, and Soot) [22].

"In the literature, one may find references of using upto 1% of nanoparticles [22]. The use of higher blends has been avoided due to stability issue (e.g. precipitation of NP from the blend)."

Mechanical and chemical methods were employed for obtaining a stable suspension of the NPs in the SPK's. Hexa-decyl tri-methyl ammonium bromide (HDTMAB) was used to inhibit the precipitation of the ceria NPs. Also mechanical vibration at 45 khz frequency was used to mix the HDTMAB, ceria NPs and SPK solution so that a stable blend is formed. X-ray diffraction (XRD) of the NPs was obtained by using powder CeO_2 samples using Rigaku MiniFlex X-ray diffractometer. Fig. 3 indicates the XRD spectra for CeO_2 NPs. Diffraction angles between 20 and 80°were used. CuKα radiation was used with wavelength equal to 0.15 nm. Peaks obtained in the diffractogram show that the synthesized NPs indeed correspond to ceria as per Joint Committee on Powder Diffraction Standards (JCPDS) standards. The average NP size was found to be 28.51 nm from Scherrer's [40] equation.

"Nanoparticles of fuel additives tend to form an aggregate owing to their large surface area and high surface reactivity. The aggregation of nanoparticles can cause sedimentation of particles and also increases the chances of clogging of the fuel injector. Typically, the size of injector diameter ranges between 125 and 200 μm, and it must be ensured that nanoparticle do not clog the atomizer. Also there are precipitation and stability issues with NP sizes greater than 50 nm. Therefore a conservative size of 30 nm particle is used" [20].

2.4. Numerical analysis

For the purpose of numerically estimating emissions, ANSYS® Fluent Software was used. The CAD model for the combustor is shown in Fig. 4a. Hexahedral meshes were created and the final mesh consisting of 6.3 million cells was chosen. A detailed study relating to mesh validation is given in [41]. Cross section of the final mesh is shown in Fig. 4c. In order to resolve the spray zone properly, the mesh is refined near the axis. Near-wall mesh is also refined to resolve the boundary layer.

The generic transport equation for fuel blended with NPs is given as:

$$\frac{\partial}{\partial t}(\rho\Phi) + div(\rho u_i \Phi) = div\{\Gamma_\Phi grad\Phi\} + S_\Phi \quad (1)$$

Specific heat capacity, molar mass and density changes occur due to addition of NP in fuels. Properties of NP additive to the fuel blends are modified as per the following equations. The approach is similar to the one adopted by Mehregan et al. [42] in their numerical study with Nanoparticles.

$$\rho_{nf} = (1-\Phi)\rho + \Phi\rho_{NP} \quad (2)$$

$$M_{nf} = (1-\Phi)M + \Phi M_{NP} \quad (3)$$

$$(\rho C_p)_{nf} = (1-\Phi)\rho C_p + \Phi\rho_{NP} C_{p,NP} \quad (4)$$

NOx is a general term for nitric oxide (NO), nitrogen dioxide (NO_2) and nitrous oxide (N_2O). The latter two occur in much smaller amounts compared to NO. NOx formation mechanism is classified as thermal, prompt, and fuel. Most of the NOx formation is by the thermal route in which nitrogen present in air is oxidized by air at high temperature during combustion. Also prompt mechanism accounts for NO alone; it is not needed for NO_2 or N_2O. The nitrogen content of jet fuel blends is

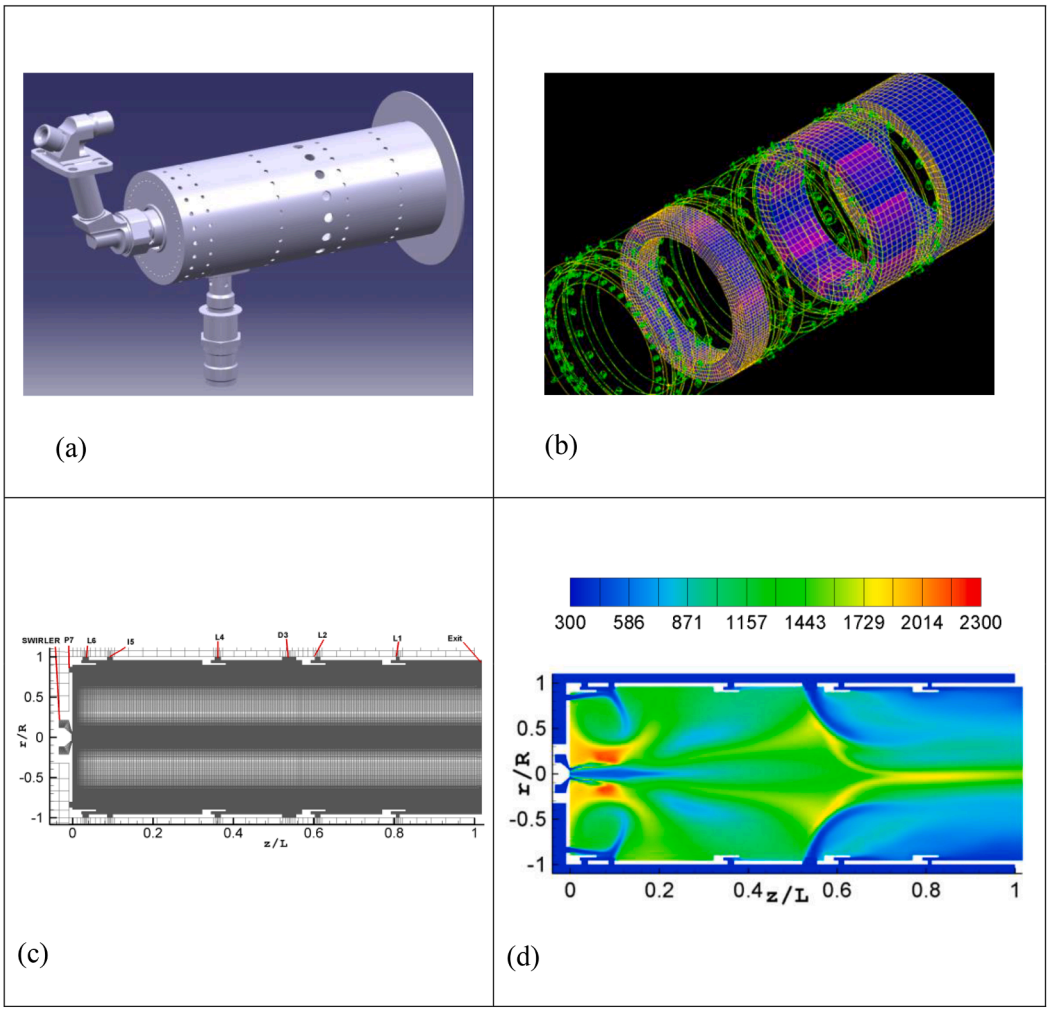


Fig. 3. Relative intensity variation with 2-theta from X-ray diffraction tests for CeO_2 NPs.

negligible and therefore not considered in the calculations. The mass transport equation for NO is written as:

$$\frac{\partial}{\partial t}(\rho Y_{NO}) + \nabla\cdot(\rho\vec{v}Y_{NO}) = \nabla\cdot(\rho D\nabla Y_{NO}) + S_{NO} \tag{5}$$

The highly temperature dependent Zeldovich reaction mechanism governs the formation of thermal NOx. The principal Zeldovich reaction steps are given below:

$$\mathbf{O} + \mathbf{N_2} \rightleftharpoons \mathbf{N} + \mathbf{NO} \tag{6}$$

$$\mathbf{N} + O_2 \rightleftharpoons O + NO \tag{7}$$

$$\mathbf{N} + OH \rightleftharpoons H + NO \tag{8}$$

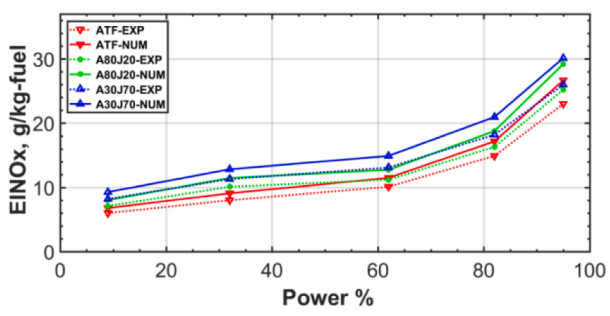
The rate coefficients for the above mentioned reactions are given from Hanson and Salimian [43].The net rate of formation of NO via the above reactions (Eqs. (6)–(8)) is given by:

$$\frac{d[NO]}{dt} = k_{f,z_1}[O][N_2] + k_{fz2}[N][O_2] + k_{f,z}[N][OH] - k_{r,21}[NO][N] - k_{r,z2}[NO][O] - k_{rz3}[NO][H] \tag{9}$$

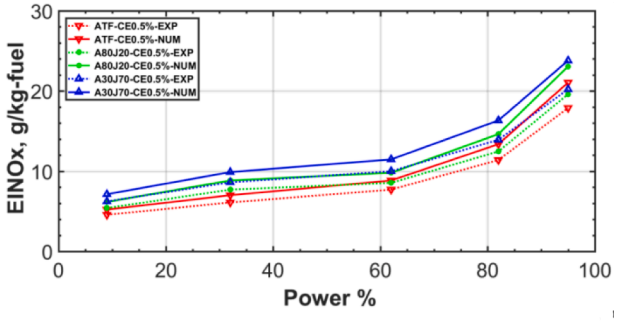
In order to calculate the formation rates of NO and N, the concentrations of O, H and OH are required. For all types of fuel–air mixtures except for very rich mixtures, the rate of oxidation of free nitrogen radicals is assumed to be equal to its formation; so a quasi-steady state is assumed for most conditions. Therefore, NO formation is considered independent of fuel type, and depends only on temperature. Following the suggestion of Zeldovich, the thermal NOx formation mechanism can be decoupled from the main combustion process by assuming equilibrium values of temperature, stable species, O-atoms and OH radicals. A more accurate partial equilibrium approach is used for determining O and OH radical concentration for the NO transport equation. Fig. 4d shows temperature contours for axial cross-section of the can-combustor. The liquid fuel is modeled by the Discrete Phase Model in ANSYS Fluent® theory guide [44]. It is a Lagrangian tracking model. The break up model is the KHRT (Kelvin-Helmholtz/Rayleigh-Taylor) hybrid model [44,45]. This model is inbuilt in ANSYS Fluent®. Spray initialization is done by mass flow rate of fuel provided at the fuel orifice modeled in the CAD geometry and mesh for numerical simulation. The evaporation model [44,46] is also built into the software ANSYS Fluent®. Species diffusion is an in-built step in the software, where the species transport equation given below is solved for the individual species.

$$\frac{\partial}{\partial t}(\rho Y_i) + \nabla\cdot(\rho\vec{v}Y_i) = -\nabla\cdot\vec{J}_i + R_i + S_i \tag{10}$$

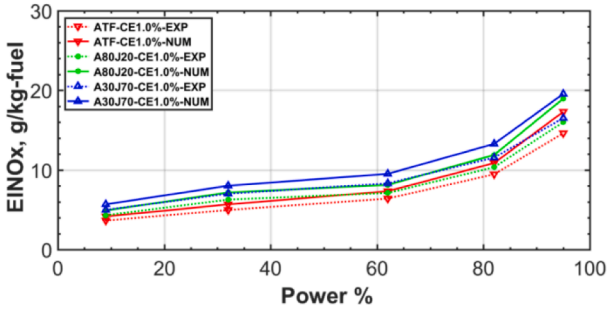
In the above equation, R_i is the net rate of production of species $_I$ by chemical reaction, and S_i is the net rate of addition from the dispersed phase (neighbouring cells), including user-defined sources. For N species, N-1 such equations will be solved. Since the mass fraction of the species must sum to unity, the Nth mass fraction is determined as one



a. NOx EI for ATF and BioJets without Ceria NP



b. NOx EI for ATF and BioJets with Ceria NP (0.5 % wt.



c. NOx EI for ATF and BioJets with Ceria NP (1.0 % wt.)

Fig. 4. (a) CAD model of can-combustor; (b) Isometric view of mesh; (c) Cross-sectional view of mesh; (d) Temperatures contours.

minus the sum of N-1 solved mass fractions. The ignition for the simulation is achieved by giving a high temperature zone near to the fuel injection in the computational domain for the first 1 s. Thereafter, the reactions within the domain could be self-sustainable. The RANS (Reynolds Averaged Navier-Stokes) model is used for modeling turbulence. For the modeling of combustion, the Non-premixed combustion model is used. In this model, the instantaneous thermochemical state is related to a conserved quantity called as the 'mixture fraction' [44,47]. This model is inbuilt in ANSYS Fluent®.

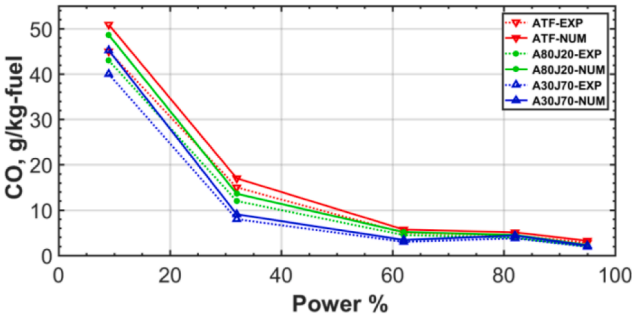
3. Results and discussion

In the following sections, the results of emissions from the exit plane of the can-combustor are presented for the nine different fuel types (Table 5) used in the study and for the five representative flight operating conditions shown in Table 1. The following subsections deal with emission results for NOx, CO, UHC and soot particle emissions, and the effect of biofuel blends and nanoparticle concentration is discussed.

3.1. NOx emissions

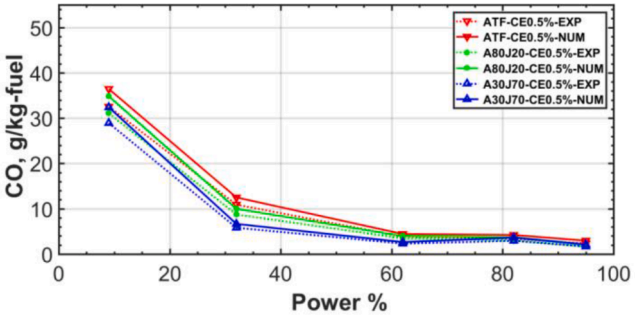
Fig. 5a shows experimental (dashed lines) and numerical results for NOx Emission Index (EI NOx) for ATF, A80J20 and A30J70. Figs. 5b and 5c show EI NOx with the same fuel blends, and with the addition of 0.5% and 1% ceria (CeO_2) NPs, respectively. For all the blends, increasing NOx with increasing power (C1 to C5; vide. Table 1) is observed. The result is consistent with the fact that NOx is primarily a function of the combustor averaged temperature and the adiabatic flame temperature [48]. Combustor averaged temperature is dependent on the overall equivalence ratio, which increases with power (Table 1). Adiabatic flame temperature depends on the primary zone equivalence ratio and heat of combustion of the fuel which is a function of H/C ratio.

Another factor that influences NOx formation is fuel atomization and evaporation. Higher viscosity results in larger atomized droplets, or



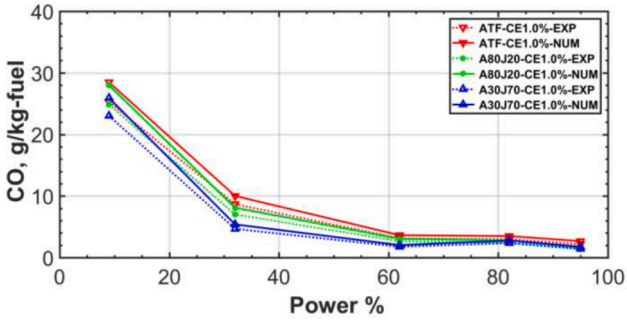
a. CO emission index for fuel blends without CeO_2

Fig. 5a. NOx EI for ATF and BioJets without Ceria NP.



b. CO emission index for fuel blends with CeO_2 (0.5% wt.)

Fig. 5b. NOx EI for ATF and BioJets with Ceria NP (0.5% wt.)



c. CO emission index for fuel blends with CeO2 (1.0% wt.)

Fig. 5c. NOx EI for ATF and BioJets with Ceria NP (1.0% wt.)

droplets with larger Sauter Mean Diameter (SMD); therefore evaporation time for the droplets would be large. Consequently, the evaporated fuel and products of combustion will spend a smaller time in the combustion chamber. Short combustor residence time results in lower NOx formation at low power. At high power conditions, due to the higher pressure, atomization occurs nearer to the atomizer and also results in smaller droplets. Therefore droplets evaporate quickly and reside in the high temperature combustor zone for a longer time as compared to low power conditions. This results in the formation of higher NOx at high power conditions [49]. There is a nonlinear increase of NOx for all fuel blends with increasing power. The aviation industry still relies on NOx correlations developed during the 1990′s. However, the various correlations are expected to perform inadequately for advanced combustor designs such as low NOx, Rich–Quench–Lean (RQL) concept, or for newer fuels or fuels with NPs as they have been developed for conventional aviation fuels.

For very low primary zone equivalence ratios (<0.4) or very lean mixtures, the NOx production has shown to be insensitive to residence time [50,51]. Therefore, fuel atomization and SMD that would depend on the fuel viscosity are unable to influence NOx formation at low equivalence ratios. Therefore, NOx formation is likely to be effected only by the H/C ratio or heat of combustion influence on flame temperature, for low power conditions. For other conditions, however, both H/C ratio and fuel atomization or SMD (indirectly, fuel viscosity and combustor pressure) should be factored in for NOx formation.

The data for EI NOx for fuel blends without NPs shows an increase with Jatropha percentage. This is due to higher heat of combustion of Jatropha, which would have a higher adiabatic flame temperature. The lower viscosity of Jatropha would result in faster atomization and evaporation, higher residence time in the combustor and therefore, higher NOx formation. For primary zone equivalence ratio<0.4 (C1), residence time is not a factor for NOx consideration, but the heating value of the fuel should be accounted for. However the impact of heating value is low at low power conditions. Hence, differences in NOx across the fuel blends are smaller for lowest power conditions (C1).

With addition of ceria to the biofuel blends, a decrease in NOx is observed for all conditions. CeO_2 (Ceria) NPs contribute towards efficient NOx reduction for size around 30 nm. CeO_2 is reduced to Ce_2O_3 when it reacts with hydrocarbons [52]. Ce_2O_3 then reduces nitrous oxide (NO) to Nitrogen. CeO_2 NPs also have higher thermal conductivity and the NPs assist in faster heat transfer, therefore quicker evaporation of fuel droplets and reduced ignition delay. This causes increase in NOx formation due to longer residence in higher temperature zones. Therefore, contribution of ceria NPs towards low NOx comes from the ability of Ce_2O_3 in reducing NO to N_2.

For higher concentration of CeO_2 (1%), further decrease in NOx emissions is observed.

Table 6 depicts percent increase or decrease in NOx for the maximum power condition. It is observed that the 36.47% reduction in NOx was observed with Jet-A + 1% CeO_2 (no Jatropha), compared to pure Jet-A.

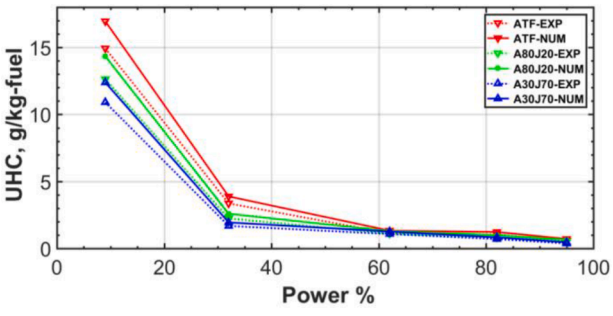
Since reduction of NO to N_2 depends on the reduction of CeO_2 via unburnt hydrocarbons, there would be a theoretical limit for NOx reduction. However at higher temperatures, there are no unburnt hydrocarbons. Also ceria promotes more complete combustion due to reduction in ignition delay and microexplosion phenomenon leading to secondary atomization; therefore there must be other pathways for NOx reduction by ceria, especially at higher power conditions, which remains an open question for research.

High pressure operations generally decrease the specific fuel consumption. However experimentation at high pressure conditions is difficult and expensive. Also, it is difficult to extrapolate low pressure results towards high pressure because results from different combustors are generally contradictory in this respect. NOx has been found to be proportional to P^n, where P is the pressure and n varies between 0.5 and 0.8 [53]. Also NOx emissions are found to be exponentially varying with temperature in the literature.

Numerical NOx estimates consistently over predict the NOx formation 15–20%. The reason may be attributed to limitations of currently used numerical models, and also due to limitations of computational resources. More processing power is needed for application of unsteady flow and/or higher order turbulence models, which is very expensive and time consuming. Furthermore, the absence of chemical kinetic mechanisms for biofuels is not available in the literature and also biofuel specifications are not standardized.

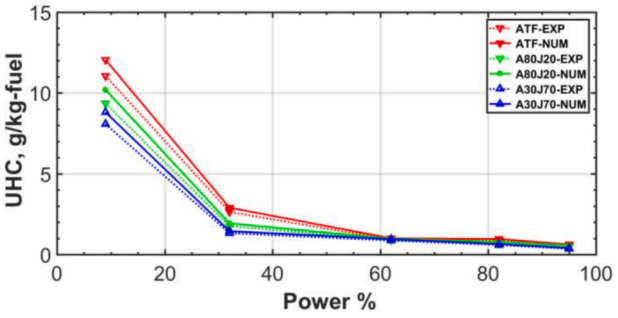
3.2. Carbon monoxide (CO) emissions and unburned hydrocarbons

CO and UHC in exhaust are indicators of incomplete combustion and reduced combustion efficiency. Mechanisms of formation of CO and UHC are the same, and they also follow similar trends, as would be evident from the results. Figs. 6a, 6b and 6c show experimental and numerical results for CO emissions. Figs. 7a, 7b and 7c show experimental and numerical results for UHC emissions. Both CO and UHC emissions reduce with increasing power, pressure or temperature.



a. UHC emission index for fuel blends without CeO_2

Fig. 6a. CO emission index for fuel blends without CeO_2.



b. UHC emission index for fuel blends with CeO_2 (0.5% wt.)

Fig. 6b. CO emission index for fuel blends with CeO_2 (0.5% wt.)

Minimum values of CO can be obtained when the primary zone equivalence ratio is slightly less (around 0.8) than stoichiometric equivalence ratio. The combustion temperature is close to maximum along with sufficient oxygen for complete oxidation of hydrocarbon and CO species.

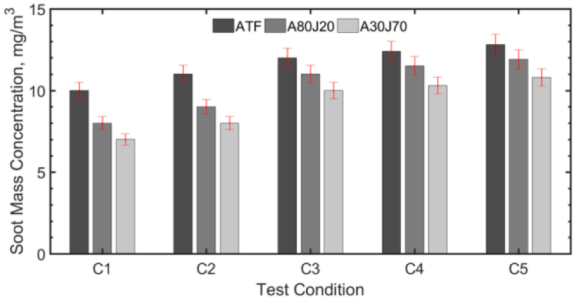
CO and UHC formation can also occur near combustor liner walls due to entrainment and cooling of fuel with liner cooling air. This becomes more likely, when atomization is insufficient, length of spray is long or spray cone angle is wide. However it has been shown that for airblast atomisers, there is no significant variation in spray cone angle [54]. Therefore, the other factor that can contribute towards above mentioned pathway would be fuel atomization. For higher pressures (or power), spray length are shorter as compared to low pressures [55]. Therefore, insufficient atomization, droplet evaporation or incomplete combustion is a characteristic of low power conditions, which can contribute towards liner-wall cooling air assisted CO and UHC.

Results consistent with the trends and factors mentioned. CO and UHC emissions are observed to decrease across the fuel blends, with increase in Jatropha content. Smaller droplets result from high temperature and pressure at high power. There is quicker atomization and evaporation and more time available for reactions to proceed towards completion. Also due to higher temperatures, completion of the oxidation reaction from CO to CO_2 becomes more probable. At higher power conditions, due to higher pressures, the spray length is reduced. This factor aids in the above mentioned quicker atomization, reaction and completion of combustion reactions. Even though the spray cone angle does not vary for air-blast atomizers, the spray length is expected to be shorter with an increase in power.

With Ceria addition to the fuel blends, there is further reduction in CO across all the fuel blends and conditions. CeO_2 aids in complete combustion of CO to CO_2. Sufficient time is available for complete combustion resulting from quicker atomization due to microexplosion phenomenon and secondary atomization. Numerical results show similar trend as the experimental results but again over predict the experimental results for all conditions and fuel combinations. Reason for this may be attributed to loss of accuracy due to improper prediction of mixing phenomenon due to lower order turbulence models.

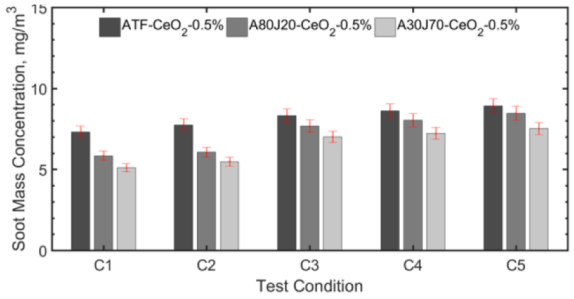
3.3. Soot emissions

Figures 8a, 8b and 8c show soot mass concentration with increasing power for the fuel blends, with and without ceria NPs. Incomplete combustion due to insufficient reaction time or low temperature or insufficient oxygen is the cause of CO or UHC. Contrary to these, soot or smoke formation occurs due to fuel-rich pockets in the combustor. In order to avoid this, sufficient air and mixing along with premixing of fuel with air are prescribed. An airblast atomizer is used in our study which inherently guarantees good atomization over the entire range of operating conditions, and also it ensures good mixing of air and fuel. Therefore atomization is not a major factor in soot formation for the combustor of study. It is seen from study of [42], that the airblast



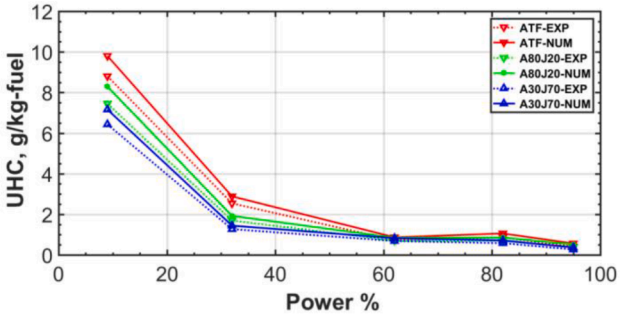
a. Soot mass concentration for fuel blends without CeO_2 NPs

Fig. 7a. UHC emission index for fuel blends without CeO_2.



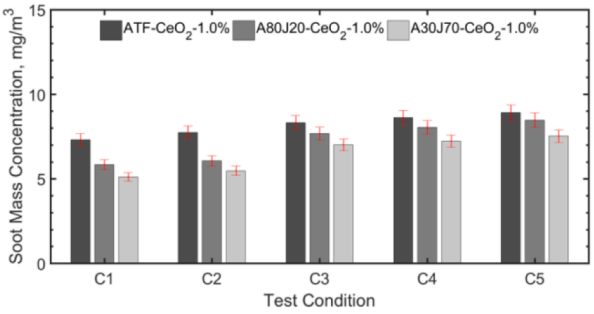
b. Soot mass concentration for fuel blends without CeO_2 (0.5% wt.)

Fig. 7b. UHC emission index for fuel blends with CeO_2 (0.5% wt.)



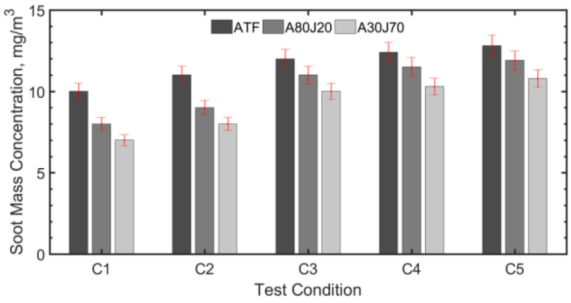
c. UHC emission index for fuel blends with CeO_2 (1.0% wt.)

Fig. 6c. CO emission index for fuel blends with CeO2 (1.0% wt.)



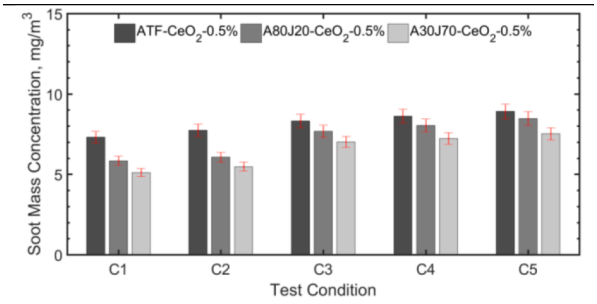
c. Soot mass concentration for fuel blends without CeO_2 (1.0% wt.)

Fig. 7c. UHC emission index for fuel blends with CeO_2 (1.0% wt.)



a. Soot mass concentration for fuel blends without CeO_2 NPs

Fig. 8a. Soot mass concentration for fuel blends without CeO_2 NPs.



b. Soot mass concentration for fuel blends without CeO_2 (0.5% wt.)

Fig. 8b. Soot mass concentration for fuel blends without CeO_2 (0.5% wt.)

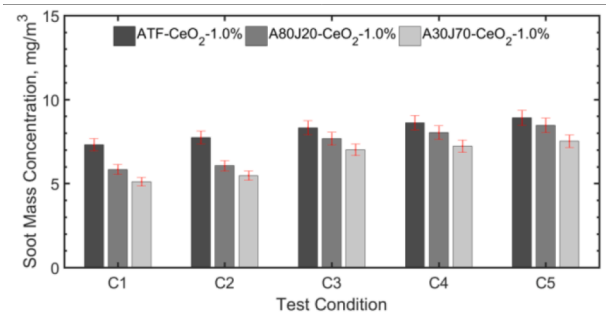


Fig. 8c. Soot mass concentration for fuel blends without CeO_2 (1.0% wt.)

atomizer has similar atomization characteristics even for biofuel blends. Therefore, there is not much variation in soot formation over the operating conditions; however the slight increase in soot at higher power conditions is due to increasingly rich primary zone mixtures (for C3, C4 and C5).

Soot formation is also influenced by chemical composition of fuel. It is well known that aromatic content and Naphthalene content are the main precursors for soot initiation and formation. However recent research suggests that naphthalenes are more effective soot precursors compared to aromatics [29]. In the Jatropha based SPK, the naphthalene content is 1.62, 1.32 and 0.61 (Table 3) for Jet-A with 20% Jatropha and Jet-A with 70% Jatropha. Also the aromatics in the last case is 25% less compared to Jet-A. Therefore, there is a decrease in soot with increasing Jatropha content and the result is consistent with literature results.

There is also a decrease in viscosity with increase in Jatropha in the fuel blends, which aids in easier atomization and smaller droplet sizes or lower SMD.

For the ceria NPs, it is observed that there is a general reduction in the soot emissions for all conditions. In addition, for higher ceria addition, there is further decrease in soot emissions. The action of CeO_2 NPs proceeds in two ways. One is surface action and the other is bulk action. Surface action is much more pronounced for lower size NPs. The phenomenon of microexplosion and catalytic activity combine to create quicker atomization and faster evaporation and reaction. Therefore, there is a decrease in soot formation due to CeO_2 addition.

Thus it is seen that CeO_2 decreases NOx, CO, UHC and also soot due to its unique properties, unlike other NPs such as Al_2O_3. Also in diesel engines, the soot or smoke formation is different because of different composition of aromatics and naphthalenes, even though the trend is the same.

4. Conclusions

Experiments were conducted for representative flight power conditions using Jet-A aviation fuel and Jet-A + Jatropha-SPK blends along with CeO_2 nanoparticle additives. NOx emissions were observed to increase with increasing power irrespective of fuel type. The observed NOx variation shows an exponential dependence on temperature and almost square root dependence on pressure. As the chemical reactions governing the formation of UHC and smoke are highly complex, most of the emission prediction correlations so far have been confined to NOx and CO. NOx emission reduction is observed with the addition of CeO_2 NPs. CeO_2 oxidizes hydrocarbons and is reduced to Ce_2O_3, which then reduces NO to N_2. 41% reduction in NOx was observed with Jet-A + 1% CeO_2 (no Jatropha), compared to pure Jet-A. However increase of up to 50% in NOx was noted for 70% Jatropha without NPs, compared to Jet-A with 1% CeO_2 NPs.

For the case of CO and UHC emissions, the factors affecting the emissions are the same, but the reaction mechanisms for UHC are much more complex. CO and UHC are found to decrease with increasing

power. Increasing the content of Jatropha and CeO_2 also aids in CO and UHC reduction. A maximum of 53% reduction in UHC and 46.7% reduction in CO was noted for 70% Jatropha along with 1% CeO_2.

Soot emissions in the study are found to increase slightly with power. A maximum of 45% decrease in soot is seen for 70% Jatropha biofuel with 1% ceria, compared to pure Jet-A. On average, 20–24% soot variation is observed from low power to high power conditions. Soot emission variations are not significant with changes in power. For higher concentrations of Jatropha and CeO_2 NPs, CO and UHC reduction occurred with an increase in power.

Numerical results for NOx, UHC and CO_2 over predict the results by 15–20%. However, the results may be taken as a first step for the redesign of the combustor. The results show that higher-order turbulence models along with parallel processing on multiple computational cores may give a better prediction.

Data Availability Statement

The required data is available in the article

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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